

LUNAR BUILD MISSION PLAN — sited  
Moon · 2026-06-03 · cut-fill balanced, optimized sequence

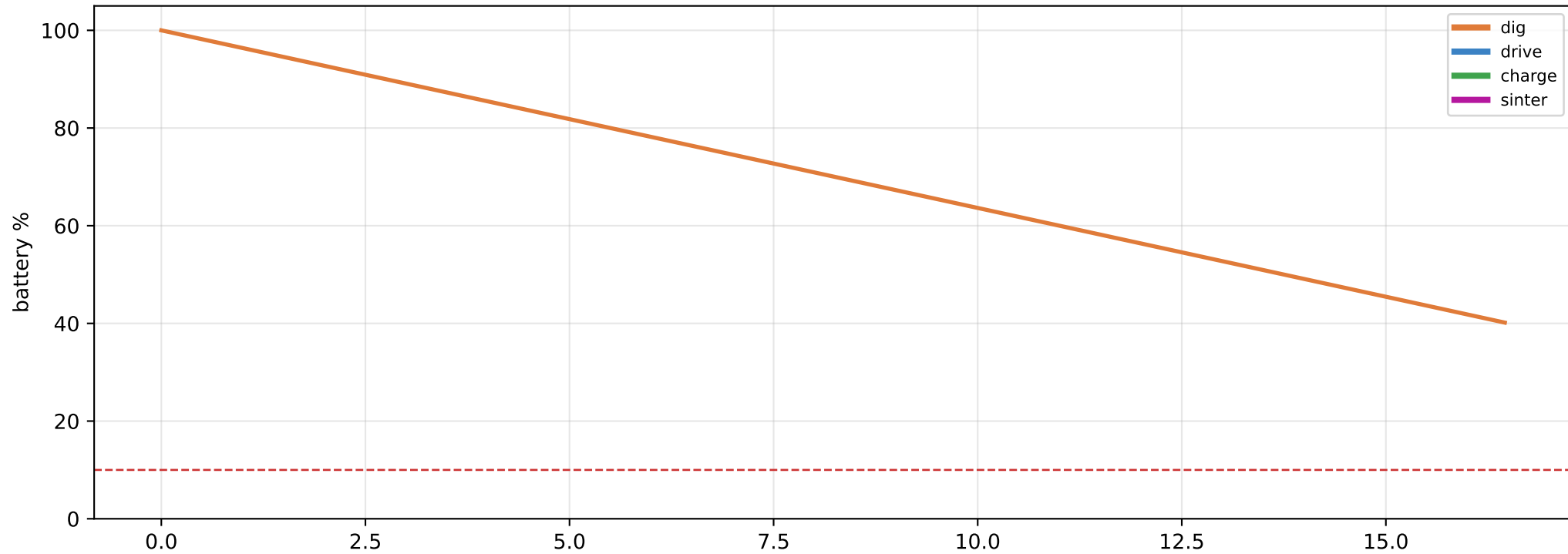
| # | Trip                | kind | Site (x,y) | Mass t | Duration | Energy (chg) |
|---|---------------------|------|------------|--------|----------|--------------|
| 1 | Excavate spoil: cut | dig  | (0,0)      | 0.69   | 16.5 h   | 0.6          |

MATERIAL BALANCE  
cut 0.7 t → fill 0.0 t · surplus(spoil) 0.7 t · deficit(import) 0.0 t · sinter 0.00 t

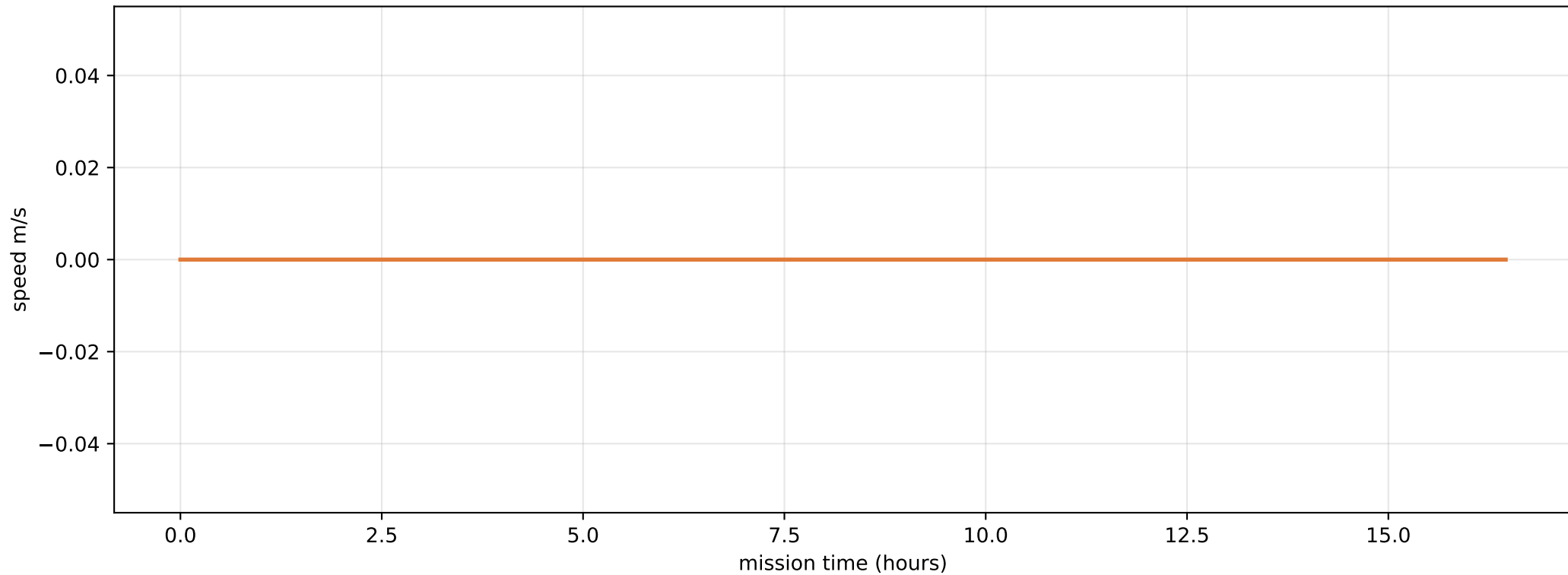
TOTALS project 16.5 h (16 h) moved 0.7 t (23 drum loads)  
energy 2.9 MJ (0.6 charges, 0 recharge stops) drive 0.00 km  
0 drum cycles; fill SENSED from motor current +/-2.6% (>half full), no load cell  
per-sortie range 24 km (slope+slip @ 20 deg; 32 km flat)  
1 lunar day ~ 30 Earth-days (354 h light)  
ConOps: 70 km + 5-10 t / 11 d -> drive ~2.0 packs vs dig ~4-9 packs: drums dominate

Cut-fill balanced (excavated material routed to nearest fill; surplus→spoil, deficit→import). Grounded: per-body density/gravity (bodies.json); IPEx — 0.30 m/s, 42 kg/hr dig, 4151 J/kg, 135 J/m (slip-adjusted on a DEM), 4.79 MJ battery, 30 kg/drum; SINTER 0.92 MJ/kg [CALIB] (~220× dig). Recharge 700 W + sinter-head 1000 W are [CALIB]. Pluggable sequencer × objective; battery-aware mid-task recharge.

Battery draw over the planned project



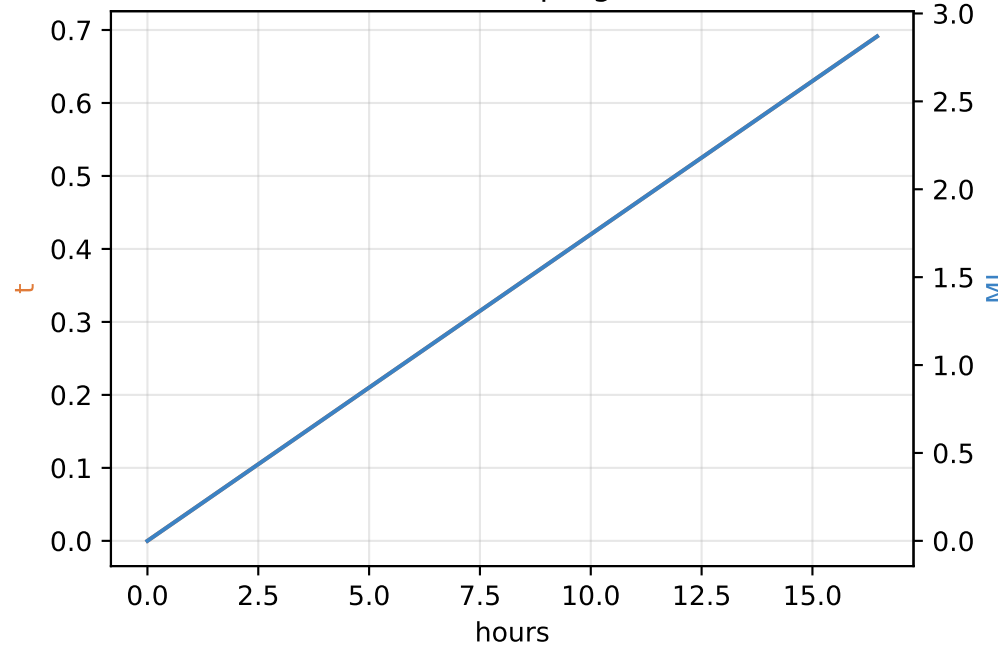
Speed profile



Site route + material flows (cut→fill)



Cumulative progress



Energy per trip (battery charges)

